Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 164–174

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Attention-Based AdaptSepCX Network for Effective Student Action Recognition in Online Learning

Arnab Dey[a,*], Anubhav Anand[a], Subhajit Samanta[a], Bijay Kumar Sah[a], Samit Biswas[a]

[a]*Dept. of Computer Science and Technology, Indian Institute of Engineering Science and Technology, Shibpur, Howrah, 711103, India*

arnabdeycs@gmail.com (Arnab Dey)., 2020csb102.anubhav@students.iiests.ac.in (Anubhav Anand)., 2020csb046.subhajit@students.iiests.ac.in (Subhajit Samanta)., bijay.csb@gmail.com (Bijay Kumar Sah)., samit@cs.iiests.ac.in (Samit Biswas).

Abstract

In the realm of online learning and distance education, the issue of inadequate supervision looms large, posing a significant obstacle. This paper delves into the challenges posed by the lack of supervision in online learning environments and proposes an innovative solution to understand and recognize students' behaviors. This study's primary objective is to detect and recognize students' actions in images captured through webcam. This task distinguishes itself from the well-established video-based student action recognition domain, which relies on temporal cues. Recognizing student actions from images intensifies the complexity of the problem. To meet this challenge, a novel deep learning model named AdaptSepCX Attention, specifically designed for student action recognition in online learning environments, is introduced. The proposed method exhibits exceptional performance with 92.73% validation accuracy on the Student Online Action Image dataset (SOAId), a carefully curated collection comprising 2029 student-centric images. The proposed model outperforms well-established models such as DenseNet121, NASNet Mobile, ConvXNet, DELVS1 and MobileNetV2 in student action recognition. Action recognition for students has broader implications beyond the online classroom. It has the potential to revolutionize educational technology, making online learning more interactive and engaging. Enabling machines to understand and respond to student actions enhances education, personalizes learning, and supports students' academic success and well-being. This research enhances the understanding of student involvement in online learning and offers an effective solution for recognizing actions from images.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Students Action Recognition; Action Recognition; Online Learning; Attention; Adaptive SeperableConvolution Network

1. Introduction

Recognizing student actions through webcams in online learning is crucial for understanding educational dynamics and enhancing teaching, learning, and student well-being. Digital learning environments, unlike traditional classrooms [1], lack real-time monitoring of emotions, concentration, and motivation. The capacity to identify and interpret student actions from the images using deep learning is a pivotal frontier in educational research, especially in today's

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.206

era of abundant image data and the demand for intelligent systems that can comprehend student behaviors. Deep learning-based frameworks for student action recognition serve as indispensable tools in education. They enable the analysis of student engagement, detection of signs of confusion or disinterest, and evaluation of online learning platform effectiveness. By decoding subtle cues from body language and gestures, these frameworks empower educators to gain profound insights into student interactions, paving the way for personalized and effective teaching strategies. Moreover, the applications of student action recognition extend beyond the classroom. In the age of data-driven decision-making, student action recognition plays a crucial role in optimizing educational content delivery by tracking student behaviors and preferences. [2] This approach tailors educational content to individual needs, ultimately enhancing the learning experience. Student action recognition through webcams in online learning represents a transformative force in education. As the potential of this technology is harnessed, the prospect emerges to create more responsive, effective, and inclusive educational experiences for students worldwide. The proposed research not only advances the field but also underscores the transformative potential of this technology in shaping the future of learning, communication, and human-machine interaction. Despite the increasing interest in student action recognition, it is noteworthy that very few studies exist in this domain. Moreover, the openly accessible datasets available for student action recognition, to date, only considered a limited subset of potential action categories.

The Adaptive Seperable ConvolutionX (AdaptSepCX) Attention Network introduced in this research represents an innovative model architecture explicitly designed to excel in recognizing Student Actions in images. This innovative network integrates adaptive separable convolutions, utilizing learnable kernel shapes for efficient feature extraction, and incorporates an attention mechanism to selectively focus on crucial regions in input data, enhancing its ability to recognize student actions in images. The AdaptSepCX Network's dual capabilities, efficient feature extraction through adaptive separable convolutions and the power of spatial attention make it particularly well-suited for tasks related to Student Actions, resulting in superior performance in action recognition scenarios. The main objective of the proposed model revolves around the classification of student action images into ten distinctive categories: *Attentive*, *Calling*, *Frustrated*, *HavingFood*, *HavingTea*, *RaisingHands*, *Sleeping*, *UsingMobile*, *Writing*, *Yawning*.

The primary contributions of this research are: (a) The development of a new image dataset, the Student Online Actions Image dataset (SOAId). (b) The successful recognition of Student Actions in images through the utilization of the novel AdaptSepCX Attention Network. (c) The proposed model is thoroughly evaluated on two closely related datasets, specifically SED and DDD. (d) Evaluation of model with Regularized cross-entropy loss

This manuscript is organized as follows: Section II highlights related studies, Section III discusses the details of the proposed method, Section IV unveils experimental findings and discussion, and lastly, the paper concludes by addressing future prospects.

2. Related Works

In this section, prior research endeavours concerning the domain of student action recognition in images and video streams. Research attention has prominently centered around examining students' behaviors within video [2][3] sequences. However, it is essential to highlight a discernible decline in research volume dedicated to delving into these aspects when transitioning from the realm of video sequences to the realm of images. Li et al. [3] proposed an attention-based relational reasoning model for Student Behavior Recognition. In another study, Rashmi et al. [4] introduced a system designed to identify various student actions within a single image frame captured by a CCTV camera. This diverse dataset encompassed actions ranging from sleeping and eating to using phones, engaging in discussions, and being actively involved in class. The researchers leveraged the YOLOv3 framework, integrating it for both object detection and the recognition of students' actions within the frame. Zhao et al. [5] have utilized deep neural networks to detect drowsy students. Delgado et al. [6] have proposed a new video dataset containing three student engagement categories namely Looking at paper, Looking at screen and Wandering. In another study, Ali et al. [7] have created a new image dataset named SUST to recognize students actions. They have utilized deep learning with YOLOv3 to recognize student actions, classified them into seven categories, and attained 97% Mean Average Precision (MAP). Cheng et al. [8] employed a Deep Convolutional Generative Adversarial Network (DCGAN) to classify student actions. Li et al. [9] have introduced a new video dataset containing fifteen student action categories. They have attained 93.1% accuracy in classifying students' actions utilizing the InceptionV3 model. Thomas et al. [10] presented a machine-learning based method aimed at assessing students' engagement within a classroom. This

method analyzed factors such as the position of head, direction of eye gaze, and facial expressions. Yu et al. [12] introduced a sophisticated deep ensemble model designed for the identification of human actions within static images. Zhang et al. [13] presented a method for identifying actions in images with a focus on reducing the requirement for extensive annotations. The prior research in Human Interaction Recognition (HIR) focused on predicting diverse human interactions from images, classifying them into fifteen types [14]. The AdaptiveDRNet model achieved a rigorous 97.20% validation accuracy when tested on the H2HId dataset. Additionally, another study addressed workout action recognition from still images [28]. Numerous studies [15, 16] [24, 26, 30] have also focused on understanding student's engagement in online learning. Hasnine et al. [25] discerned students' engagement in online education by extracting emotions. Shou et al. [17] aims to enhance student action recognition in class time-series images using an enhanced Asynchronous Interaction Aggregation (AIA) model. Mo et al. [18] have introduced a Multitask heatmap network to categorize student behaviors from still images. Zheng et al. [27] introduced a novel dataset encompassing five distinct classroom behaviors: writing, listening, sleeping, reading, and talking, and the authors have classified these behaviors within the dataset. Yang et al. [29] have directed their attention towards classifying student behaviors in the physical classroom into six categories, namely reading, writing, hand-raising, using a phone, leaning over the table and bowing the head. Zhang et al. [31] have introduced a novel approach to hand-raising gesture recognition in the classroom, leveraging a dilated convolution network. Abdallah et al. [19] utilized VGG-16 based network to recognize Student Behaviors in the classroom. Buono et al. [20] have accessed student engagement from facial behavior in online learning. It is observed that there is a limited consideration of student action recognition from images.

Standard image datasets, namely ActClassD [18] and SED [21], have played a pivotal role in advancing student action recognition. Nonetheless, these openly accessible datasets have taken into account only a subset of the possible action categories. Initially, the ActClassD [18], and SED [21] datasets focused on seven and six distinct student engagement categories, respectively. In an effort to enrich available resources for researchers and practitioners in this domain, the Student Online Action Image dataset (SOAId) has been introduced, encompassing ten diverse categories of student actions. The main goal is to provide the researchers with valuable resources to develop and evaluate precise models for classifying and recognizing student actions in images. This dataset is poised to improve the ability of models to generalize and exhibit real-world relevance in recognizing student actions within online learning environments.

3. Proposed Method

The approach began by gathering images of students' online actions and proceeded with data augmentation, resizing, and normalization processes. This meticulously prepared dataset was then partitioned into train, validation, and test sets. Data augmentation techniques were utilized to augment the training data, augmenting its effectiveness. The proposed model was pivotal in training and extracting essential image features. An insightful representation of the proposed methodology can be observed in Fig. 1.

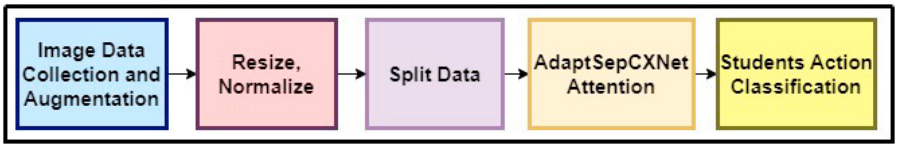


Fig. 1: Workflow representation

In response to the demand for resource-efficient research methods, this study presents the AdaptSepCX Attention Network, a solution for recognizing diverse student actions in images. The following sections delve into the specifics, covering: a) Preparing the SOAId dataset, b) Implementation details, c) The architecture of *AdaptSepCX Attention Network*, and d) Loss function.

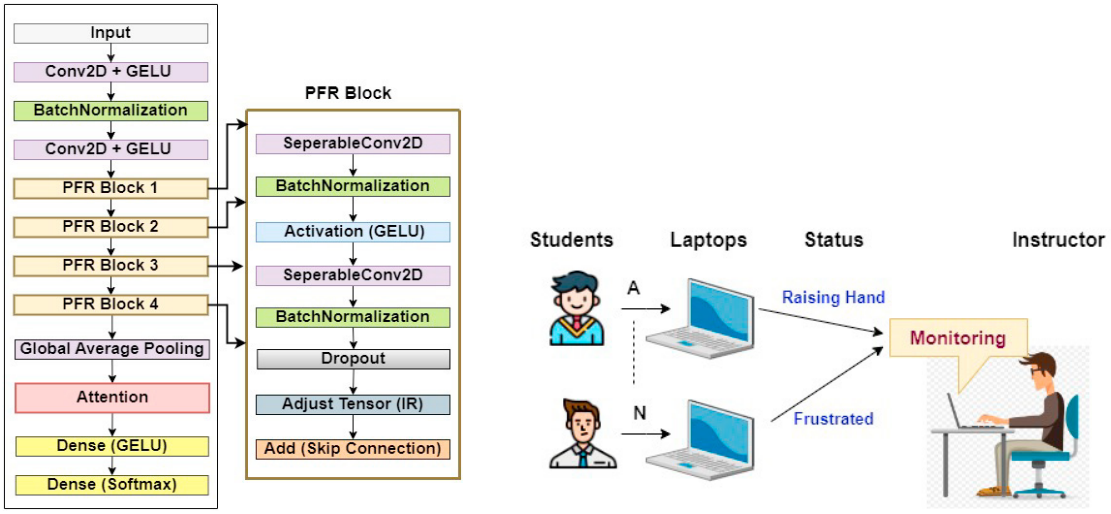
3.1. Dataset Preparation:

The SOAId dataset, which focuses on Student Online Actions, was carefully curated. It underwent augmentation and resizing to a standardized 224 x 224 resolution for all RGB images. This dataset comprises ten distinct student

action categories. Data augmentation is strategically employed to enhance the performance of the AdaptSepCX Attention model. This augmentation involves expanding the training dataset size and mitigating overfitting through the incorporation of random transformations, such as zoom, contrast adjustments, rotations, and horizontal flips. These transformations introduce diversity into the training data, exposing the model to a range of data distribution characteristics. Moreover, the meticulous division of the SOAId dataset into three distinct subsets— comprising training (61%), validation (20%), and testing (19%)—significantly enhances the robustness of the model's training and evaluation processes.

3.2. Implementation Details:

For instance, consider the context of students participating in online classes, where they are seated in front of their laptops. In this scenario, the images of students can be captured using their laptop webcam at every five-minute interval during the online classes. The proposed AdaptSepCX Attention Network, which has been trained specifically for recognizing and classifying various student actions, then processed these captured images. By integrating this model into the monitoring system, instructors gain a powerful tool to track and assess the status and actions of students during online classes. This approach enhances the instructor's ability to maintain a productive and engaging learning environment, as they can promptly identify any signs of disengagement or issues that might require attention. Consequently, the AdaptSepCX Attention model, combined with webcam image capture, becomes an effective aid for instructors in monitoring student participation and ensuring the success of online learning sessions. The implementation framework is depicted in Fig. 2b.



(a) Proposed AdaptSepCX Attention model

(b) Implementation Framework

Fig. 2: Proposed AdaptSepCX Attention Model and Implementation Framework

3.3. AdaptSepCX Attention Network: Model Architecture

The proposed AdaptSepCX model with scalar dot product attention is structured in a progressive and hierarchical manner, designed to extract increasingly complex features from input images. The diagram illustrating the proposed Adaptive Seperable Convoulution-X (AdaptSepCX) Attention model for implementing student action recognition is illustrated in Fig. 2a. This diagram encapsulates the model's architecture, providing a concise reference for its implementation and design intricacies. The model begins with an initial Convolutional Layer with GELU activation, which reduces spatial dimensions while extracting low-level features. Then, another Convolution layer with GELU activation is incorporated. Following this, four Progressive Feature Refinement (PFR) Blocks are utilized. Each PFR Block consists of two Separable Convolution Blocks, incorporating two consecutive SeparableConv2D layers with batch normalization, GELU activation, dropout and Skip Connection. After the PFR blocks, a Global Average Pooling (GAP) Layer is employed to reduce spatial dimensions to 1x1 while retaining channel information. The proposed

model further includes a scalar dot product attention mechanism, which processes the global average pooled feature map to compute attention scores and scales the feature map accordingly. Finally, the dense output layer employs the softmax activation function for classification into ten student action classes. This comprehensive architecture offers adaptability and complexity, with the number of layers and filters easily adjustable to suit specific tasks, making it well-suited for image classification challenges. The proposed model has 381,514 million trainable parameters.

Here's a detailed description of the proposed AdaptSepCX Attention model architecture, broken down into individual blocks:

Input Layer: The input to the model consists of images with dimensions of 224x224 pixels and three color channels representing red, green, and blue (RGB).

Initial Convolution Layer: The model initiates with an initial Convolutional Layer, designed to process input images characterized by a (224, 224, 3) shape. This layer utilizes a Conv2D operation incorporating the Gaussian Error Linear Unit (GELU) activation function, accompanied by 32 filters, each featuring a kernel size of (3, 3) and a stride of (2, 2). Following the convolution operation, the next step involves the application of batch normalization to standardize the activations. Then, another Conv2D layer with GELU activation function and 64 filters, each having a kernel size of (3, 3) and a stride of (2, 2), is incorporated. The primary purpose of this layer is to extract essential low-level features from the input images, concurrently diminishing spatial dimensions to optimize the data for subsequent processing stages.

The mathematical representation of a Conv2D layer followed by a GELU activation can be expressed as a composition of functions. Let $\mathbf{X}$ be the input tensor, $\mathbf{W}$ be the weight tensor of the convolutional layer, and $\mathbf{b}$ be the bias vector. The Conv2D operation can be written as illustrated in Eqn. 1. Here, $*$ denotes the convolution operation, and σ represents the element-wise activation function applied after the convolution.

$$Conv2d(\mathbf{X}) = \sigma(\mathbf{W} * \mathbf{X} + \mathbf{b}) \tag{1}$$

The GELU activation function is defined as illustrated in Eqn. 2. Here, x is the result of the Conv2D operation.

$$GELU(x) = \frac{1}{2}x\left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}\left(x + 0.044715x^3\right)\right)\right) \tag{2}$$

Progressive Feature Refinement (PFR) Block: The model then progresses through a series of four SeparableConv Blocks. These PFR blocks are a crucial part of the model and consist of custom cells that progressively refine and enhance features. Each custom cell within this block follows a consistent pattern. Each block consists of two Separable convolutional layers with 3x3 kernels and padding set to 'same'. This is followed by the sequential application of batch normalization, GELU activation, a dropout layer, an optional Tensor adjustment step (if needed), and finally, a skip connection for enhanced model connectivity. Batch normalization and GELU activation functions are applied after each convolution to ensure stable training. Adjust the Residual Tensor (IR) step ensures that the number of filters in the residual tensor matches the number of filters in the output tensor from the second separable convolution. If they don't match, a 1x1 separable convolution is applied to the residual tensor to adjust the number of filters. The final output of the block is obtained by adding the output of the second separable convolution and the adjusted residual tensor. This creates a skip connection (Residual Connection), which combines low-level and high-level features.

The Separable convolution 2D is mathematically represented using Eqn. 3. This equation combines both the horizontal and vertical convolutions into a single expression for separable convolution. The separable convolution of a 2D input image I with separable kernels K_x and K_y can be represented as:

$$(I * K_x * K_y)[i, j] = \sum_{q=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} I[i - q, j - r] \cdot K_x[q] \cdot K_y[r] \cdot \delta[q, r] \tag{3}$$

Here, $(I * K_x * K_y)[i, j]$ denotes the result of the separable convolution at position (i, j), $I[i, j]$ signifies the pixel value of the input image at position (i, j), $K_x[q]$ denotes the values of the horizontal kernel at position q, $K_y[r]$ represents the values of the vertical kernel at position r and $\delta[q, r]$ represents the Kronecker delta function, which is 1 when $q = r$ and 0 otherwise.

Moreover, a skip connection connects PFR Block 1 to PFR Block 2 to facilitate the flow of information. In the first two blocks, referred to as PFR Block 1 and PFR Block 2, both blocks use 128 filters for feature extraction. PFR

Block 2 builds upon the hierarchical representation of features established in PFR Block 1 and further enhances the model's ability to capture intricate details within the data. PFR Block 2 and PFR Block 3 share a similar structure with PFR Block 1 but differ in the number of filters used in its convolutional layers, which is increased to 128. PFR Block 3 closely resembles PFR Block 2 but employs an even higher number of filters, set to 256. This increased capacity enables PFR Block 3 to continue amplifying the feature hierarchy, extracting abstract features that may be crucial for the model's classification task. Following the first two blocks, the model advances to the second and third blocks, PFR Block 3 and PFR Block 4, where each block employs 256 filters for feature extraction. Similar to the first and second PFR blocks, these blocks utilize separable convolutional layers, batch normalization, and residual connections.

This PFR Block is an integral part of the model and helps in feature extraction and information flow within the network. It allows for efficient training and captures meaningful patterns in the data.

Global Average Pooling (GAP) Layer: Following the four PFR blocks, a GAP Layer is introduced. This layer performs global average pooling on the feature maps obtained from the previous blocks. The process reduces the spatial dimensions of the feature maps to 1x1 while retaining channel information. This step prepares the model's output for classification by providing a condensed representation of the learned features.

Attention Mechanism: In this proposed model, an advanced scalar dot product attention mechanism is employed to assign weights to the feature vectors produced by the GAP Layer. The following elucidates the working mechanism:

Scalar Dot Product Computation: After the global average pooling operation, a lambda layer is used to compute the sum of values along the channel axis, effectively creating a scalar for each example in the batch. This step is represented by Eqn. 4. Here, C denotes the number of feature channels and x_i represents the value of the i-th channel in the global average pooled feature tensor.

$$Att_{Scores} = \sum_{i=1}^{C} x_i \tag{4}$$

Softmax Activation for Attention Weights: The computed attention scores (Att_{Scores}) undergo a transformation through a softmax activation function, yielding attention weights ($Att_{Weights}$) as illustrated in Eqn. 5 that collectively add up to 1. These weights effectively signify the importance attributed to each individual feature channel in the process.

$$Att_{Weights} = Softmax(Att_{Scores}) \tag{5}$$

Element-wise Multiplication with Feature Vectors: Finally, the attention weights are applied element-wise to the original feature vectors from the Global Average Pooling (GAP) Layer as illustrated in Eqn. 6. This operation adjusts the importance of each channel in the feature vectors based on the computed attention scores. This step effectively amplifies or attenuates the contribution of each channel's information in the final feature representation.

$$Att_{WeightedFeatures} = Att_{Weights} \cdot GAP_{Features} \tag{6}$$

Thus, this advanced scalar dot product attention mechanism computes attention scores by summing feature values across channels, applying a softmax activation to obtain attention weights, and then using these weights to adjust the feature vectors. The attention mechanism allows the model to focus on the most relevant information within the feature vectors and adaptively weigh the contributions of different channels during inference.

Output Layer: The model culminates with two Dense layers, one employing 64 units with GELU activation and the other dense layer employing ten units with a softmax activation function. The last Dense layer is crucial and is tasked with generating the model's predictions, serving as the ultimate decision-maker in the model's output. It generates a probability distribution over the specified number of output classes. By using the features learned throughout the preceding layers, the output layer performs the actual classification, assigning probabilities to each class based on the input data.

Thus, the proposed model architecture is organized into blocks; each custom PFR block includes separable convolution blocks and skip connections. These blocks progressively extract and refine features.

3.4. Loss Function:

To address the multi-class nature inherent in the student action recognition problem, the selected methodology involves the application of the Regularized categorical cross-entropy loss (RC_{Loss}) function [14]. This function serves

as a reliable measure for quantifying the model's error reduction objective during its training process. The formulation of the RC_{Loss} is depicted in Eqn. 7. In this equation, y_{tp} represents the actual probability distribution (one-hot encoded labels) for class membership, y_p represents the anticipated probability distribution (output from the model) for class membership, and M is set to ten, indicating the number of student action classes in the dataset. The RC_{Loss} function incorporates an additional regularization term, represented by the second term in the expression.

$$RC_{Loss} = -[\sum_{1}^{M=10} y_{tp}^{(k)} \cdot \log(y_{pp}^{(k)})] + \sum_{1}^{M=10} (y_{tp}^{(k)} - y_{pp}^{(k)})^2 \tag{7}$$

The Adam optimizer is employed, utilizing an initial learning rate of 0.001, to proficiently minimize the error function associated with the proposed AdaptSepCX Attention model.

4. Experimental Results

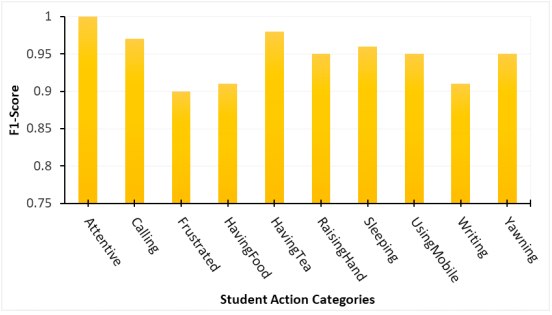
The research is performed in Google Colab, employing Python for resource-efficient processing without GPU usage. Here, the specifics of the collected SOAId dataset are presented, encompassing available datasets, a rigorous performance assessment, and in-depth results analysis.

4.1. Dataset Details

Numerous publicly available datasets focus on action recognition, such as SED [21] and DDD [22] datasets. However, the ActClassD dataset [18], which pertains to student actions, is not publicly available. The SED dataset features images of student engagement scenarios, while DDD data showcases driver actions. Due to limited student action data, the driver state dataset is utilized for the analysis, given its relevance and shared attributes. The Students Online Action Image dataset (SOAId) [1] have been created for this purpose. SOAId is unbiased, featuring a variety of student actions across ten categories with 2029 images. The SOAId data has been collected from various sources such as Google Images, 500px, Shutterstock and YouTube. The meticulous organization included labelling images of distinct Student action classes and segregating them into specific directories. Each directory was systematically named to align with the corresponding student action depicted in the images. The student action categories considered are *Attentive*, *Calling*, *Frustrated*, *HavingFood*, *HavingTea*, *RaisingHands*, *Sleeping*, *UsingMobile*, *Writing*, *Yawning*. Some sample images of the SOAId data are given in Fig. 3a.



(a) SOAId Data Sample



(b) Classification Performance of SOAId

Fig. 3: SOAId Data Sample and Classification performance of the Proposed model.

[1] **SOAId Dataset Link**: https://sites.google.com/view/student-actions/

4.2. Performance Evaluation

The evaluation of this research's efficacy encompasses a diverse set of metrics, including train and validation accuracy, AUC-score, F1-score, and the confusion matrix. The proposed model is configured with a 224x224 pixel input size, a modest batch size of 5, and is subjected to 24 epochs of training in a ten-class classification scenario. It harnesses a 3x3 kernel, initializes learning with a rate of 0.001, and refines its performance using the Adam optimizer. The chosen optimization strategy is coupled with Regularized Categorical Cross-entropy loss and accuracy as the principal evaluation metric. The proposed model achieved a noteworthy 96.60% training accuracy and 92.73% validation accuracy.

Classification Performance: The proposed model's effectiveness in classifying student actions is vividly portrayed in Fig. 3b, revealing the F1-scores for the SOAId dataset. Remarkably, the majority of student action categories in the SOAId dataset exhibit impressive F1 scores, surpassing 0.95. Nonetheless, a few categories, such as "HavingFood" (0.91), "Frustrated" (0.90), and "Writing" (0.91), display slightly lower but still commendable F1 scores. It underscores the exceptional classification capabilities of the AdaptSepCX Attention model on the SOAId dataset.

The study evaluated various established deep learning models on the SOAId dataset, as depicted in Table 1. These models underwent comprehensive training from scratch exclusively with the SOAId dataset. The dataset was partitioned into training (61%), validation (20%), and testing (19%) sets for the evaluation process. The analysis, focusing on validation accuracy, F1-Score, and AUC (Area Under the Curve) scores, provides valuable insights into their performance. The Proposed Model emerged as the top performer, with a validation (Val.) accuracy of 92.73%. Other models, including DenseNet121, ConvXNet, DELVS1[12] and NASNet Mobile [11], also displayed robust performance, achieving validation accuracies of 91.30%, 90.76%, 91.72% and 92.32%, respectively. In terms of the F1-Score, the Proposed Model achieved the highest score of 0.95, followed by the NASNet Mobile [11], DenseNet121 and DELVS1 [12] models with a score of 0.93, 0.93 and 0.92, respectively on the SOAId dataset.

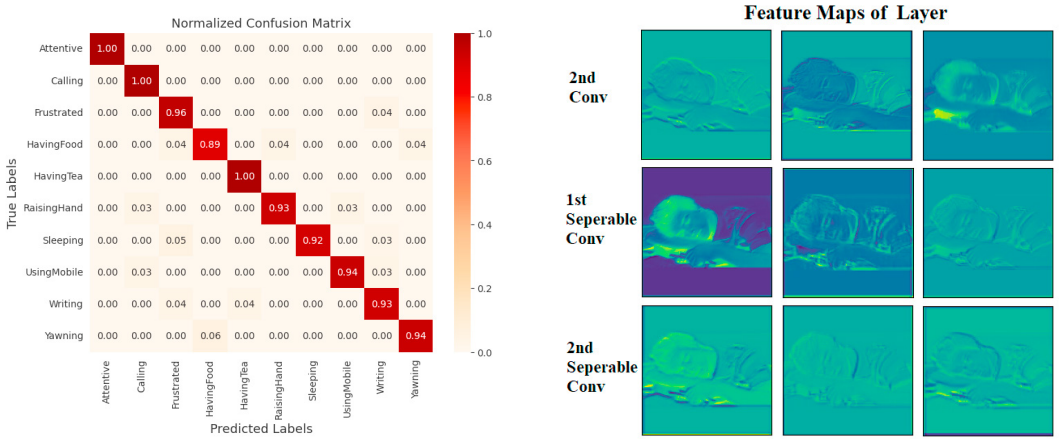
Table 1: Evaluating the Performance of various benchmark models on the SOAId Dataset

Model Considered	Train Accuracy	Val. Accuracy	F1-Score	AUC
VGG-16	94.21%	88.27%	0.89	0.94
ResNet50	96.08%	90.54%	0.90	0.95
MobileNetv2	96.38%	91.41%	0.91	0.96
DenseNet121	97.61%	91.30%	0.93	0.97
ConvXNet	96.28%	90.76%	0.91	0.96
DELVS1[12]	96.48%	91.72%	0.92	0.97
NASNet Mobile [11]	96.70%	92.32%	0.93	0.98
Proposed Model	**96.60%**	**92.73%**	**0.95**	**0.99**

When evaluating deep learning models on the SOAId dataset using AUC scores, the effectiveness of each model was carefully assessed. The results underscored the superiority of the Proposed Model, which demonstrated an exceptional AUC score of 0.99, highlighting its remarkable ability to distinguish between diverse classes. Following closely, the NASNet Mobile [11] and DELVS1 [12] models demonstrated robust performance, achieving AUC scores of 0.98 and 0.97, respectively. In contrast, other models displayed relatively lower performance, with the VGG-16 model recording the least AUC score at 0.94. These results underscore the outstanding performance of both the proposed model and NASNet Mobile in accurately discerning positive and negative instances within the SOAId dataset.

Confusion Matrix and Feature Map: The visual representation depicted in Fig. 4a offers a comprehensive insight into the normalized confusion matrix pertaining to the SOAId dataset, thereby providing a nuanced understanding of the model's performance across various student actions. The matrix serves as a graphical representation of the model's predictive accuracy by delineating correct and incorrect classifications. Upon careful examination of the matrix, it becomes evident that the diagonal elements exclusively encapsulate instances of accurate predictions, signifying the model's proficiency in correctly identifying and categorizing specific student actions within the dataset. However, a discerning analysis of the normalized confusion matrix reveals that actions, namely "Writing", "HavingFood", "RaisingHand" and "Sleeping", exhibit a relatively lower normalized score than 0.94.

The filter outputs from a 'Sleeping' student action image are visually presented in Fig. 4b. It shows the generated feature maps of the proposed model's 2nd Conv2D, 1st Seperable Conv2D (PFR Block 1) and 2nd Seperable Conv2D (PFR Block 1) layers of the proposed model. The layer numbering corresponds to the sequence outlined in the layered architecture of the proposed AdaptSepCX model (See Fig. 2a). The feature maps generated at each of these distinct layers encapsulate hierarchical representations of the input image.



(a) Confusion Matrix of SOAId dataset

(b) Feature map of 'Sleeping' student action

Fig. 4: Confusion Matrix and Generated Feature map.

The AdaptSepCX Attention model, as proposed, has demonstrated precise predictions for all image samples of Student's online actions during the subsequent evaluation, as visually represented in Fig. 5.

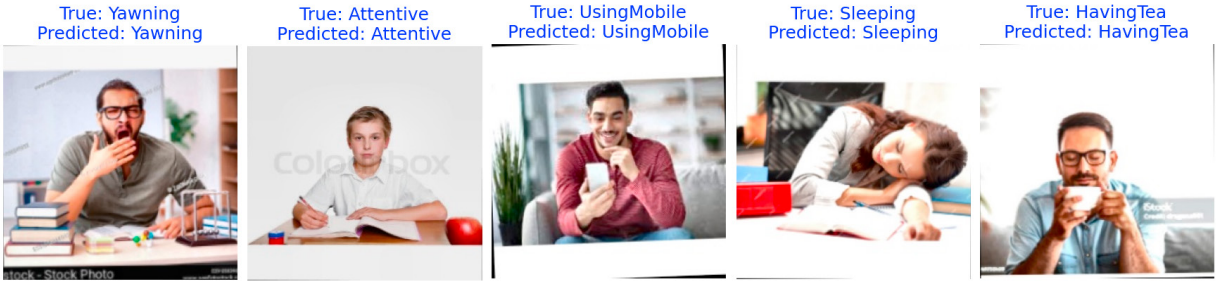


Fig. 5: Test Result of Proposed AdaptSepCX Attention Model

The proposed method underwent extensive testing across multiple established CNN models, as illustrated in Table 2. These models were trained entirely from scratch with the related datasets, utilizing pertinent datasets such as SED [21], DDD [22], and the collected SOAId dataset. To facilitate evaluation, the datasets were partitioned into training, validation, and testing subsets in 61%, 20% and 19%, respectively. Among the various models evaluated, the proposed model demonstrated superior performance, achieving an outstanding accuracy rate of 98.38% on the SED dataset. It outperformed other models such as NASNet Mobile [11] and DELVS1 [12], which attained validation accuracies of 97.63% and 97.20% on SED data, respectively. In comparison, DenseNet121 and MobileNetv2 achieved accuracies of 96.28% and 94.54% on the SED dataset. For the DDD dataset, DELVS1 [12], NASNet Mobile [11], DenseNet121, and I-AlexNet [23] models achieved validation accuracies of 84.76%, 85.35%, 85.43%, and 83.37%, respectively. In contrast, ResNet50 and MobileNetv2 models achieved accuracies of 80.20% and 85.10%, respectively, on the DDD dataset. Remarkably, the proposed approach exhibited impressive validation accuracies of 86.46% on the DDD dataset [22] and 92.73% on the collected SOAId dataset. This underscores the exceptional capability of the proposed model in accurately identifying student actions, showcasing remarkable overall accuracy.

Table 2: Comparative Assessment of various models for Student Action Recognition

Method(s)	SED (%)	DDD (%)	SOAId (%)	T. Params
VGG-16	95.20	81.56	88.27	134.7 M
ResNet50	94.56	80.20	90.54	26 M
MobileNetv2	94.54	85.10	91.41	3.50 M
NASNet Mobile [11]	97.63	85.35	92.32	5.30 M
DenseNet121	96.28	85.43	91.30	7.98 M
DELVS1 [12]	97.20	84.76	91.72	>140 M
I-AlexNet [23]	95.74	83.37	90.56	0.586 M
Proposed Model	**98.38**	**86.46**	**92.73**	**0.381 M**

When examining these models based on the count of trainable parameters (T. params), it is worth highlighting that the suggested model stands out for its relatively streamlined architecture, encompassing only 0.381 million (M) T. params. In contrast, certain benchmark models showcase substantially higher quantities of trainable parameters. For instance, the DELVS1 Model [12] surpasses 140 million T. params, ResNet50 boasts 26 million T. params, DenseNet121 model reaches 7.98 million, VGG-16 employs 134.7 million T. params, MobileNetv2 employs 3.50 million T. params, I-AlexNet [23] employs 0.586 million and NASNet Mobile [11] model employs 5.30 million T. parameters. The key strength of the proposed model lies in its capacity to attain higher accuracy while upholding notably fewer trainable parameters compared to these widely recognized benchmark models. This quality holds particular relevance in real-world scenarios where memory constraints carry the utmost significance.

5. Conclusion and Future Scope

Student Action Recognition in Online Learning is a swiftly evolving domain, propelled by deep learning methodologies and bearing profound importance across diverse sectors. Employing this model to distinguish intricate student behaviors and interactions from static visual data unlocks opportunities in online education, surveillance, and human-machine interactions. This study introduces an innovative approach, fusing attention mechanisms with separable convolution and skip connections, amplifying the model's capability to automatically extract pertinent features within images. This advancement is poised to significantly enhance the precision and efficacy of identifying complex student actions, rendering it indispensable in online education and human-computer interaction. Principal contributions of this investigation encompass the creation of a comprehensive Student Online Actions Recognition dataset (SOAId) with meticulous labelling, the deployment of a specialized AdaptSepCX Attention model for discerning student actions in images, and a thorough model evaluation on well-established datasets, including SED and DDD, employing a range of performance metrics. Notably, the proposed model's efficiency enables it to be deployed on resource-constrained devices, requiring only a modest number of trainable parameters. This study highlights the significant impact of deep learning in enhancing the understanding of student behaviors and actions in online learning derived from visual data while concurrently emphasizing its extensive applicability across diverse domains. The future scope of this research involves expanding the dataset with a greater variety of student action categories and extending the work on recognizing student actions in video streams, leveraging a lightweight model.

References

[1] Yang, B., Yao, Z., Lu, H., Zhou, Y., and Xu, J. (2020). "In-classroom learning analytics based on student behavior, topic and teaching characteristic mining." *Pattern Recognition Letters*, **129**, 224-231.

[2] Lin, F. C., Ngo, H. H., Dow, C. R., Lam, K. H., and Le, H. L. (2021). "Student behavior recognition system for the classroom environment based on skeleton pose estimation and person detection." *Sensors*, **21(16)**, 5314.

[3] Li, Y., Qi, X., Saudagar, A. K. J., Badshah, A. M., Muhammad, K., and Liu, S. (2023). "Student behavior recognition for interaction detection in the classroom environment." *Image and Vision Computing*, 104726.
[4] Rashmi, M., Ashwin, T. S., and Guddeti, R. M. R. (2021). "Surveillance video analysis for student action recognition and localization inside computer laboratories of a smart campus." *Multimedia Tools and Applications*, **80**, 2907-2929.
[5] Zhao, G., Liu, S., Wang, Q., Hu, T., Chen, Y., Lin, L., and Zhao, D. (2018). "Deep convolutional neural network for drowsy student state detection." *Concurrency and Computation: Practice and Experience*, **30(23)**, e4457.
[6] Delgado, K., Origgi, J.M., Hasanpoor, T., Yu, H., Allessio, D., Arroyo, I., Lee, W., Betke, M., Woolf, B. and Bargal, S.A., 2021. "Student engagement dataset." In Proceedings of the IEEE/CVF International Conference on Computer Vision.
[7] Ali, M. Y., Zhang, X. D., and Harun-Ar-Rashid, M. (2020). "Student activities detection of SUST using YOLOv3 on deep learning." *Indonesian Journal of Electrical Engineering and Informatics (IJEEI)*, **8(4)**, 757-769.
[8] Cheng, Y., Dai, Z., Ji, Y., Li, S., Jia, Z., Hirota, K., and Dai, Y. (2020). "Student action recognition based on deep convolutional generative adversarial network." *2020 Chinese Control And Decision Conference (CCDC)*, 128-133.
[9] Li, X., Wang, M., Zeng, W., and Lu, W. (2019). "A students' action recognition database in smart classroom." *2019 14th International Conference on Computer Science and Education (ICCSE)*, 523-527. IEEE.
[10] Thomas, C., and Jayagopi, D. B. (2017). "Predicting student engagement in classrooms using facial behavioral cues." *Proceedings of the 1st ACM SIGCHI international workshop on multimodal interaction for education*, 33-40.
[11] Zoph, B., Vasudevan, V., Shlens, J., and Le, Q. V. (2018). "Learning transferable architectures for scalable image recognition." *Proceedings of the IEEE conference on computer vision and pattern recognition*, 8697-8710.
[12] Yu, X., Zhang, Z., Wu, L., Pang, W., Chen, H., Yu, Z., and Li, B. (2020). "Deep ensemble learning for human action recognition in still images." *Complexity*, 2020, 1-23.
[13] Zhang, Y., Cheng, L., Wu, J., Cai, J., Do, M. N., and Lu, J. (2016). "Action recognition in still images with minimum annotation efforts." *IEEE Transactions on Image Processing*, **25(11)**, 5479-5490.
[14] Dey, A., Biswas, S., and Le, D. N. (2023). "Recognition of Human Interactions in Still Images using AdaptiveDRNet with Multi-level Attention." *International Journal of Advanced Computer Science and Applications*, **14(10)**.
[15] Liu, H., Liu, Y., Zhang, R., and Wu, X. (2021). "Student behavior recognition from heterogeneous view perception in class based on 3-D multiscale residual dense network for the analysis of case teaching." *Frontiers in neurorobotics*, **15**, 675827.
[16] Ruiz, N., Yu, H., Allessio, D.A., Jalal, M., Joshi, A., Murray, T., Magee, J.J., Delgado, K.M., Ablavsky, V., Sclaroff, S. and Arroyo, I., 2022. "ATL-BP: a student engagement dataset and model for affect transfer learning for behavior prediction." *IEEE Transactions on Biometrics, Behavior, and Identity Science.*
[17] Shou, Z., Yan, M., Wen, H., Liu, J., Mo, J., and Zhang, H. (2023). "Research on Students' Action Behavior Recognition Method Based on Classroom Time-Series Images." *Applied Sciences*, **13(18)**, 10426.
[18] Mo, J., Zhu, R., Yuan, H., Shou, Z., and Chen, L. (2023). "Student behavior recognition based on multitask learning." *Multimedia Tools and Applications*, **82(12)**, 19091-19108.
[19] Abdallah, T. B., Elleuch, I., and Guermazi, R. (2021). "Student behavior recognition in classroom using deep transfer learning with VGG-16." *Procedia Computer Science*, **192**, 951-960.
[20] Buono, P., De Carolis, B., D'Errico, F., Macchiarulo, N., and Palestra, G. (2023). "Assessing student engagement from facial behavior in on-line learning." *Multimedia Tools and Applications*, **82(9)**, 12859-12877.
[21] Student Engagement Dataset (SED). [Online Available] www.kaggle.com/datasets/joyee19/studentengagement
[22] Nasri, I., Karrouchi, M., Snoussi, H., Kassmi, K., and Messaoudi, A. (2022). "Detection and prediction of driver drowsiness for the prevention of road accidents using deep neural networks techniques." *WITS 2020: Proceedings of the 6th International Conference on Wireless Technologies, Embedded, and Intelligent Systems*, 57-64. Springer Singapore.
[23] Li, S., Wang, L., Li, J., and Yao, Y. (2021). "Image classification algorithm based on improved AlexNet." *Journal of Physics: Conference Series*, 1813(1), 012051. IOP Publishing.
[24] Su, K., Li, X., Zhou, C., and Chen, X. (2021). "Learning behaviour recognition based on multi-object image in single viewpoint." *Personal and Ubiquitous Computing*, **25**, 1081-1090.
[25] Hasnine, M. N., Bui, H. T., Tran, T. T. T., Nguyen, H. T., Akçapınar, G., and Ueda, H. (2021). "Students' emotion extraction and visualization for engagement detection in online learning." *Procedia Computer Science*, **192**, 3423-3431.
[26] Hu, M., Wei, Y., Li, M., Yao, H., Deng, W., Tong, M., and Liu, Q. (2022). "Bimodal learning engagement recognition from videos in the classroom." *Sensors*, **22(16)**, 5932.
[27] Zheng, Z., Liang, G., Luo, H., and Yin, H. (2022). "Attention assessment based on multi-view classroom behaviour recognition." *IET Computer Vision.*
[28] Dey, A., Dutta, A. and Biswas, S. (2023). "WorkoutNet: A Deep Learning Model for the Recognition of Workout Actions from Still Images." *In 2023 3rd International Conference on Intelligent Technologies (CONIT)*. IEEE, pp. 1-8.
[29] Yang, F., and Wang, T. (2023). "SCB-Dataset3: A Benchmark for Detecting Student Classroom Behavior." arXiv preprint arXiv:2310.02522.
[30] Zhao, J., Zhu, H., and Niu, L. (2023). "BiTNet: A lightweight object detection network for real-time classroom behavior recognition with transformer and bi-directional pyramid network." *Journal of King Saud University-Computer and Information Sciences*, **35(8)**, 101670.
[31] Zhang, G., Wang, L., Wang, L., and Chen, Z. (2023). "Hand-raising gesture detection in classroom with spatial context augmentation and dilated convolution." *Computers & Graphics*, **110**, 151-161.